

Cost-benefit analysis technique

cost-benefit analysis (CBA) is a technique used to evaluate the potential benefits and costs associated with developing and implementing a software project or feature

A **cost-benefit analysis** is the process of comparing the projected or estimated costs and benefits (or opportunities) associated with a project decision to determine whether it makes sense from a business perspective.

Cost-benefit analysis (CBA) is an essential technique in software development, helping teams make informed decisions about projects, features, or investments

- **Identify Costs and Benefits:** Firstly, all costs associated with the development and maintenance of the software are identified. This includes direct costs like salaries, infrastructure costs, licensing fees, etc., as well as indirect costs like opportunity costs, training expenses, etc. Similarly, potential benefits such as increased revenue, cost savings, improved efficiency, etc., are identified.
- **Calculate ROI and Payback Period:** With quantified costs and benefits, the Return on Investment (ROI) and Payback Period are calculated. ROI is typically calculated as $(\text{Net Benefits} / \text{Costs}) * 100$, where $\text{Net Benefits} = \text{Total Benefits} - \text{Total Costs}$. The Payback Period is the time it takes for the project to recoup its initial investment
- **3.Risk Assessment:** Risks associated with the project are identified and assessed. These could include technical risks, market risks, regulatory risks, etc. Risk mitigation strategies are then developed to address these risks.
- **Sensitivity Analysis:** Sensitivity analysis is performed to understand how changes in key assumptions or variables (such as project duration, costs, benefits, etc.) affect the project's outcomes. This helps in understanding the robustness.

Return of Investment (ROI):

Return on Investment (ROI) lets you determine the profit or loss generated when you invest money in a company or asset/venture

Return on investment is a performance metric that assesses an investment's effectiveness or profitability. One can also use it to compare the efficiency of investments. ROI quantifies the amount of return on a particular investment in relation to its cost.

Calculation of Return on Investment

$ROI = \text{Net profit or loss} / \text{cost of investment} * 100$

$ROI = \text{present value} - \text{cost of investment} / \text{cost of investment} * 100$

For example, Person A buys shares for INR 1,000 and, one year later, sells them for INR 1,500. In this instance, the investment's net profit (current value - cost) is INR 500 (INR 1,500 - INR 1,000), and the return on investment is:

$$\begin{aligned} ROI &= (500/1000) * 100 \\ &= 50\% \end{aligned}$$

Payback Period

Payback period analysis is a method used to determine the relative time value of a project. It measures the time that a project will take to return its initial investment. The total time taken by the project to return the initial investment made for the project is called the payback period.

payback period analysis helps in calculating and managing the risk. Hence, it is an essential factor for decision-making.

$\text{Payback Period} = \text{Initial Investment} / \text{Annual Payback}$

For example, imagine a company invests 200,000 in new manufacturing equipment which results in a positive cash flow of 50,000 per year

$\text{Payback Period} = 200,000 / 50,000$

In this case, the payback period would be 4 years because 200,000 divided by 50,000 is 4. As mentioned, the shorter the payback period, the more desirable an investment is

Feasibility Report:

A feasibility report is a report that evaluates a set of proposed project paths or solutions to determine. The person who prepares a feasibility report evaluates the feasibility of different solutions and then chooses their recommendation for the best solution. They then present the feasibility report to their company and make their recommendation.

A feasibility report studies a situation (for example, a problem or opportunity) and a plan for doing something about it, and then determines whether that plan is “feasible

Note

A feasibility report aims to determine the feasibility of solutions or project paths and choose the best option. The feasibility report serves to break down different approaches to a problem or project and help readers understand the feasibility of each approach. Based on the evaluation outlined in the report, readers can decide whether to take the report's recommendation of the best approach. This thorough analysis of different approaches can help companies make the best possible decisions on projects and problems.

System Proposal and its components:

A system proposal is a formal document that outlines a proposed solution to address a specific problem or opportunity within an organization. It serves as a comprehensive plan for initiating a project to develop or implement a new system or enhance an existing one. Here are the typical components of a system proposal:

The proposed system outlines the objectives, scope, and key features that the project aims to achieve. It provides a blueprint. In the context of projects, a proposed system refers to a new or improved system that is being proposed as a solution to address a particular problem or meet specific requirements

1. **Introduction:** This section provides an overview of the proposal, including a brief description of the problem or opportunity, the purpose of the proposal, and its importance to the organization.
2. **Executive Summary:** A concise summary of the entire proposal, highlighting the key points such as the problem statement, proposed solution, objectives, and expected benefits.
3. **Background:** This section provides background information on the current system or processes, including any challenges or limitations that the proposed system aims to address. It may also include relevant industry trends or regulatory requirements.
4. **Problem Statement:** A clear and concise statement of the problem or opportunity that the proposed system intends to solve or capitalize on. It defines the scope and context of the project.
5. **Objectives:** The specific goals and objectives of the proposed system, outlining what the system aims to achieve and how it aligns with the organization's overall strategic objectives.